

Enzymes.

* Definition - Enzymes are biological catalyst produced by living tissues. They are proteins (except RNA).

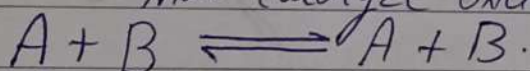
* Classification :- Classification as per International Union of Biochemistry and Molecular Biology [IUBMB]

- the seven classes as per IUBMB are as follows:-

- 1) EC - 1 : Oxidoreductase
- 2) EC - 2 : Transferase
- 3) EC - 3 : Hydrolase
- 4) EC - 4 : Lyase
- 5) EC - 5 : Isomerase
- 6) EC - 6 : Ligase
- 7) EC - 7 : Translocases.

1) EC : 1 - Oxidoreductase :

- That catalyze oxidation reduction.



(Red) (ox) (ox) (Red)

- If the reaction involves a direct participation of O_2 the enzyme termed oxidases.

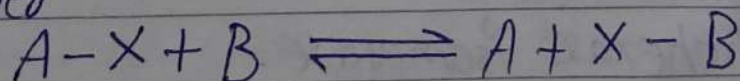
- If the reaction involves the transfer of Hydrogen the enzyme are known as dehydrogenase

eg:- lactate dehydrogenase (LDH)

G-6-P dehydrogenase

2) EC : 2 - Transferase :

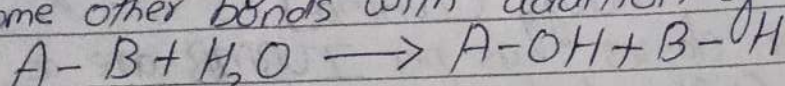
- That catalyze the transfer of group such as amino, carboxyl, methyl or phosphoryl from one molecule to another



eg. Aminotransferase or transaminase
Kinase eg. Hexokinase

3) EC: 3 - Hydrolase:

- This catalyze the cleavage of C-O, C-N, C-C & some other bonds with addition of H_2O

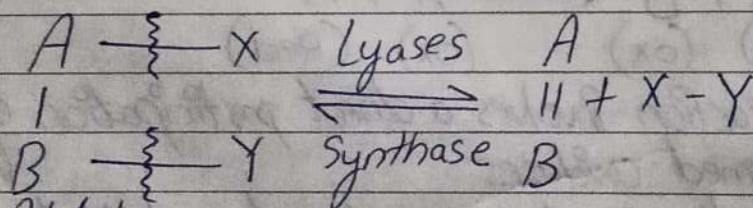


eg: Acid phosphatase

All digestive enzymes like, amylase, pepsin, trypsin, chymotrypsin.

4) EC: 4 - Lyase:

- The catalyze cleavage of C-O, C-C & C-N bonds
- Other than hydrolysis or oxidation with double bonds
- Catalyze the reverse reaction by addition of group double bond
- In case of reverse reaction synthase is used in the name.

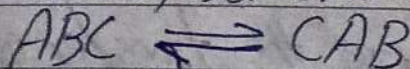


eg: Aldolase

Carbonic anhydrase

5) EC: 5 - Isomerase:

- Intramolecular rearrangement in a molecule called epimerases, isomerases or mutases.



eg: phosphoglucomutase,

Triphosphate isomerase

phosphohexase isomerase.

6) E6: 6-Ligases:

- Catalyze the joining of 2 molecules couple with hydrolysis of ATP.

eg: $A + B + \text{ATP} \rightarrow AB + \text{ADP} + \text{P}_i$
 Glutamine Synthetase, DNA ligases, pyruvate carboxylase.

7) EC: 7-Translocases [A new EC Class].

- Catalyze movement of ions or molecules cross the membrane

a) (carnitine acyl) carnitine translocase:- Responsible for fatty acid from cytosol into mitochondria.

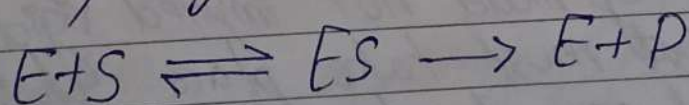
b) Ornithine translocase:- Responsible for transporting ornithine from cytosol into mitochondria.

* Mechanism of enzyme action.

1) Formation of enzyme + Substrate [ES] complex is first step of catalysis.

2) Substrate bound with or through multiple non-covalent interaction at active site of enzyme form Enzyme-Substrate complex.

3) which subsequently convert to product & free the enzyme

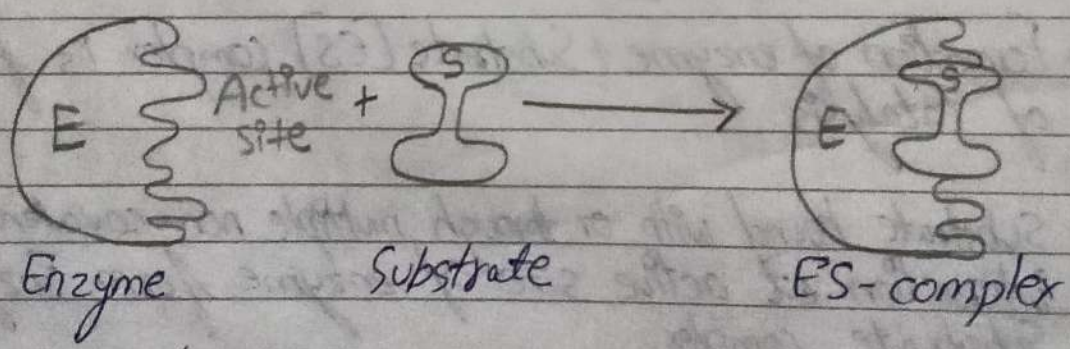


4) Two models of substrate enzyme specificity explained as follows:-

- a) Lock & key model / Rigid template model of Emil Fischer
 b) Induced fit model / Hand-in-glove model of Daniel E. Koshland

a) Lock & key model :-

- This model was proposed by Emil Fischer in 1890.
- Its name indicate that because in this model substrate fits into the active site is such a way that a key fits into a lock.
- This model has been useful in understanding some enzyme can bind only a specific substrate but not bind to another compound with different structure.
 e.g.:- most enzymes in carbohydrate metabolism can bind to D-Isomers of hexoses but cannot bind to L-Isomers of same
- This model explain all mechanisms but do not explain changes in enzyme activity in presence of allosteric modulators



b) Induced fit model / hand-in-glove model of Daniel E. Koshland :-

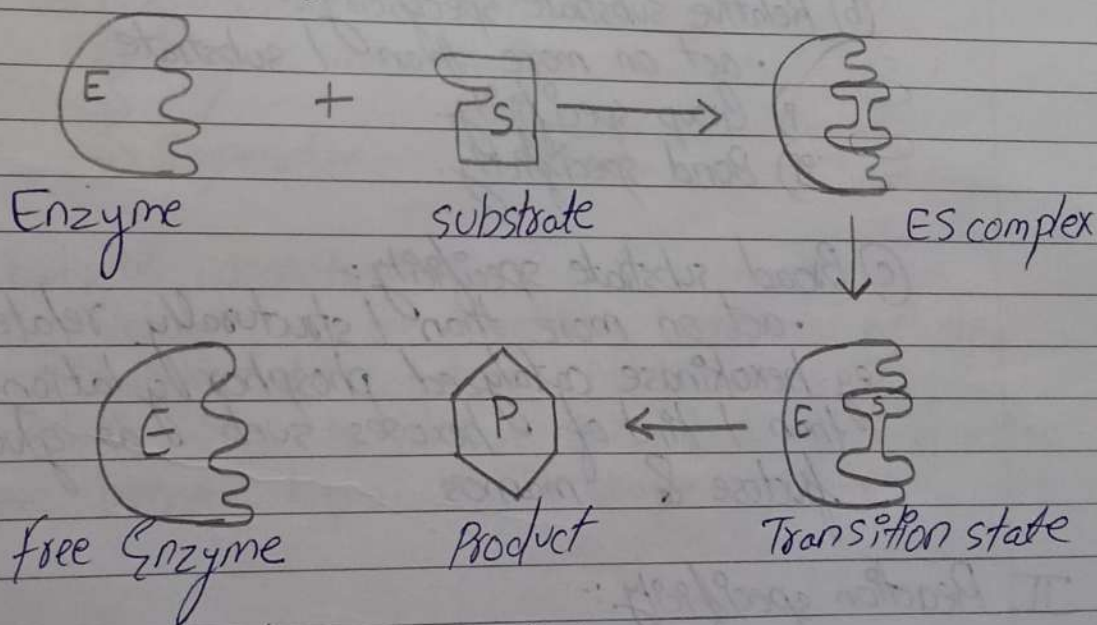
- Lock & key model not implied rigidity of enzyme active site that must take place during catalysis.
- This model accounts for both these aspects of enzyme catalysis
- In this theory, or model postulated that the enzymes

are flexible & shapes of the active sites can be modified by the binding of the substrate.

- The substrate induces a conformational change in the enzyme in the same manner, in which placing a hand (substrate) into a glove (sub enzymes) induces changes in the glove's shape therefore this model is also known as Hand-in-glove model.

- Intrinsic binding energy due to the substrate-enzyme interaction made available for the transformation of the substrate into product.

- This model described more accurately the specificity of substrate binding.



* Specificity of enzyme activity:-

Specificity is the ability of an enzyme to discriminate between two competing substrate

- enzymes are highly specific in both
 - 1) The reaction catalyzed.

2) choice of substrates.

- Specificity makes it possible for a number of enzyme to co-exist in the cell without interfering in each other's actions.

• Types of specificity:-

Substrate specificity	Reaction Specificity	stereo specificity.
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(a) Absolute substrate specificity:-

• act only 1 substrate & catalyze 1 reaction
e.g. glucokinase, lactase.

(b) Relative substrate specificity:-

• act on more than 1 substrate

i) Group specificity.

ii) Bond specificity.

(c) Broad substrate specificity:-

• act on more than 1 structurally related substrate
e.g. hexokinase catalyzed phosphorylation of more than 1 kind of hexoses such as glucose, fructose & mannose.

II Reaction specificity:-

- enzyme is specific to a particular reaction & catalyzed only 1 type of Reaction.

eg:- pyruvate undergoes several reaction.

Each reaction catalyzed by separate enzyme.

III Stereo specificity:-

- enzyme shows specificity towards stereoisomers i.e. act only 1 type isomer.
- eg 1) L-Lactate dehydrogenase :-
 - act only on L-lactic acid & not D-lactic acid.
- 2) D-glucose oxidase
 - act only on D-glucose & not L-glucose.
- 3) Salivary α -amylase.
 - act on the α -1,4 glycoside linkage & is inactive on β -1,4-Gl.

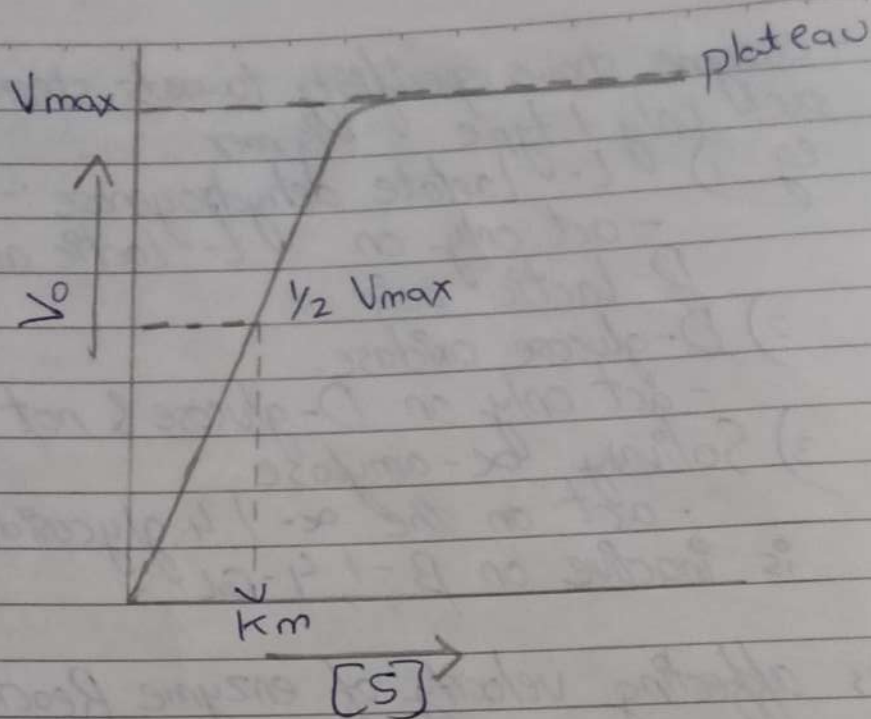
* Factors affecting velocity of enzyme Reaction.

Various factors that affect enzyme activity:-

- 1) Substrate concentration.
- 2) Enzyme concentration.
- 3) pH
- 4) Temperature.

1) Substrate concentration:-

- Velocity of the reaction increases as the concentration of substrate is increases.
- Relationship is almost linear but later, reaction curve become hyperbolic in shape.



where,

V_0 = Initial velocity, V_{max} = maximum velocity.
 K_m = Michaelis constant, S = Substrate concentration.

i.e. $K_m = V_{max}/2$

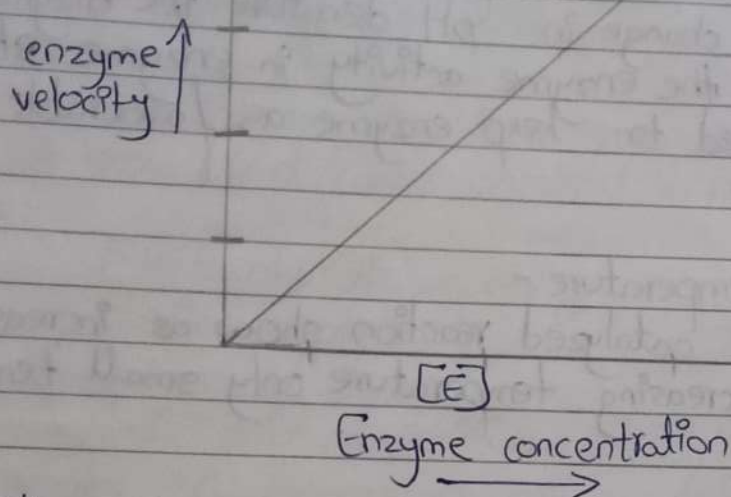
• K_m has 2 significance one is provide amount of substrate concentration required to catalysis & another is affinity of enzyme for its substrate.

* High K_m = low affinity & weak binding &
 Low K_m = High affinity & strong binding.

2) Effect of enzyme concentration:-

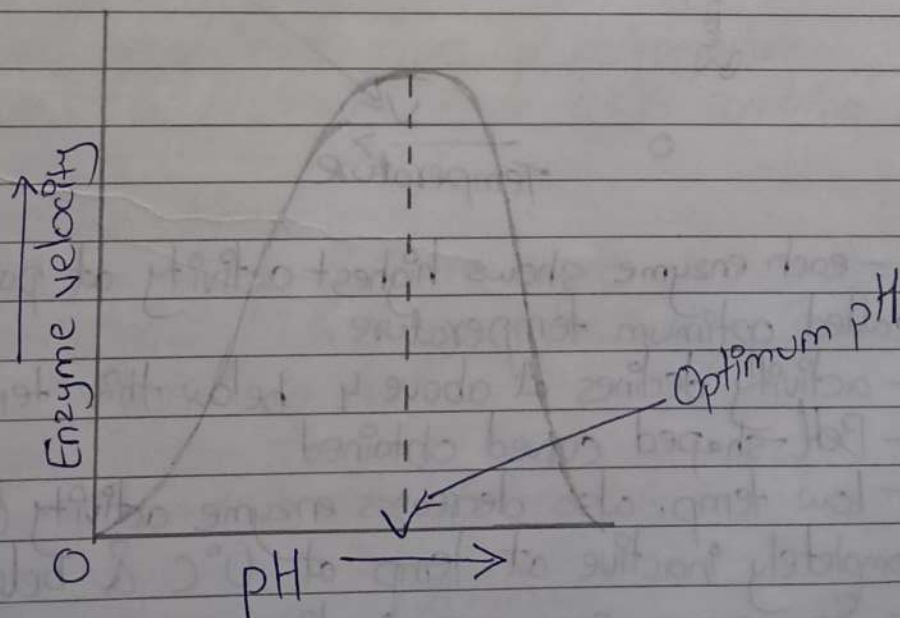
- Velocity of reaction is directly proportional to amount of enzyme present.

- substrate must be present as concentration sufficient to produce catalysis.



3) Effect of pH :-

- each enzyme has an optimum pH.
 - pH at which enzyme activity is maximal.
 - Below or above this pH enzyme activity is decreased.
- eg. pepsin = 1.2 pH & trypsin = 8.0 pH.



- Bell-shaped curve is obtained when enzyme velocity is plotted.

- changes in pH can alter the following:-

1) Ionization state of amino acid.

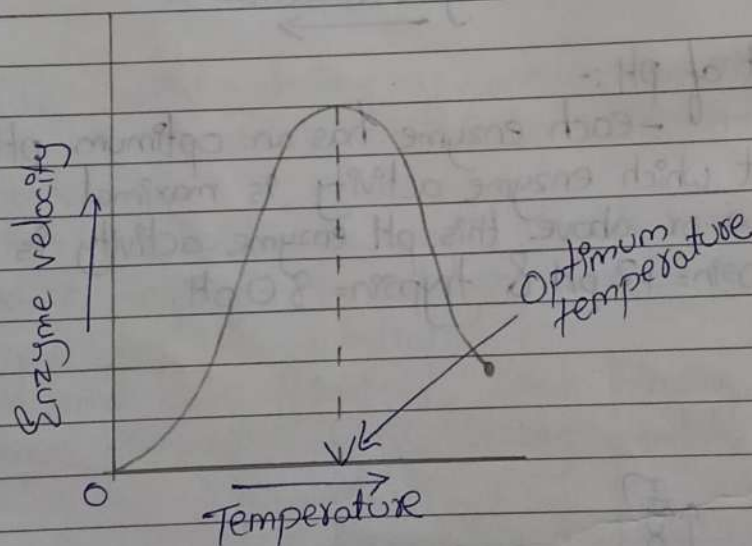
2) Ionization state of substrate.

3) Drastic change in pH denature the enzyme protein.

- as pH effect the enzyme activity in enzyme studies buffer are used to keep enzyme as favourable H^+ ion concentration.

4) Effect of Temperature:-

- Enzyme catalyzed reaction show as increase in rate with increasing temperature only small temperature range.



- each enzyme shows highest activity at particular temperature called optimum temperature.

- activity declines at above & below this temperature.

- Bell-shaped curved obtained.

- low temp. also decreases enzyme activity & enzyme maybe completely inactive at temp. at $0^{\circ}C$ & below.

- So many enzymes in tissues preserved for months by storing at $-20^{\circ}C$ or $-70^{\circ}C$.

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* Isoenzyme

- Isoenzymes are multiple forms of same enzyme.
- different isoenzymes catalyze the same chemical reaction but differ in their primary structure & kinetic properties.
- Not all enzymes has isoenzymes
- This show different chemical & physical properties such as:-

- 1) Electrophoretic mobility.
- 2) Kinetic properties.
- 3) Amino acid sequence
- 4) Amino acid composition.

Examples of Isoenzymes are as follows.

1) Lactate dehydrogenase (LDH):-

Catalyze oxidation of L-lactate to pyruvate having 5- Isoenzymes.

- (a) LDH₁ is fast moving fraction. (b) LDH₂ (c) LDH₃ ✓
 LDH₁ is fast moving fraction. (d) LDH₄ (e) LDH₅
 LDH₅ is slow moving.

- LDH made up of two types of polypeptide
 M [Muscle] & H [Heart] type & 5 combine.

2) Creatinine Kinase :-

This are dimer that are made up two type of polypeptide chains M & B type, generating 3 isoenzymes.

CK₁ = Present in brain

CK₂ = Cardiac tissue

CK₃ = Present in skeletal muscle.

3) Alkaline phosphate (ALP)

4) Acid phosphatase (ACP)

5) Amylase

6) Hexokinase

* Diagnostic & therapeutics uses & enzymes:-

Certain enzymes are used for diagnosis of diseases.

- 1) Lactate Dehydrogenase [LDH]
 - elevation of LDH₁ & LDH₂ occurs within 24-48hrs after myocardial infarction.
 - predominant elevation of LDH₂ & LDH₃ occurs in Leukemia.
 - Elevation of LDH₅ occurs after damage to the liver & skeletal muscle.

2) Creatinine Kinase [CK]:-

- CK₁, elevated in neonates particularly in damage brain.
- CK₂ in blood characteristic of damage of heart tissue from myocardial infarction.
- CK₃ level elevation in serum occurs in all type of dystrophies & myopathies.

3) Alanine Transaminase [ALT]:-

- Normal ALT plasma value for adult is 10-40 U/L.
- ALT level is elevated in liver diseases, jaundice & cirrhosis of liver.

4) Aspartate Transaminase [AST]:-

- The plasma AST normal value for adults is 10-30 U/L.
- Increased AST level occurs after myocardial infarction.
- Moderate elevated in liver diseases.

5) Alkaline phosphate [ALP]:-

- Normal serum level for adult is 3-13 kA units/dL.
- It is elevated in certain bone & liver disease.
- Very high level may be noticed in obstructive jaundice, bone diseases, rickets, osteomalacia, carcinoma of bone.

5) Acid phosphatase [ACP]:-

- Normal serum value for ACP is 0.5-4 kA units/dL.
- Prostatic acid phosphatase enzyme is useful for diagnosis & prognosis of prostate cancer.
- ACP is imp tumor marker.

6) Amylase:-

- Normal serum value is 50-120 U/L.
- activity of serum amylase is $\uparrow$ in acute pancreatitis, found in urine.
- Increase in serum level are also seen in chronic pancreatitis, mumps & obstructive of pancreatic duct.

7) Prostate specific Antigen :- [PSA]
elevated blood level occurs in prostate cancer.

* Co factor / Coenzyme / Activator. -

- Some enzyme requires an additional non-protein component for its optimum activity. this additional component is called cofactor.
- This cofactor may be either loosely or tightly bound to protein portion of the enzymes.

- This cofactor may be :-
- 1) Organic compound is coenzyme.
- 2) Inorganic ions i.e. activators.

- * Coenzyme :- act as transient carriers of specific functional groups.
- most coenzymes are derived from vitamins
 - A coenzyme that is bond to enzyme protein called as prosthetic group.
 - A complete catalytical active enzyme called Holoenzyme.
 - A enzyme without its cofactor called apoenzyme

$$\text{Holoenzyme} = \text{Apoenzyme} + \text{Cofactor}$$

eg:-

Vitamin	Coenzyme	Function as coenzyme
1) Thiamine (B ₁)	TPP [Thiamine pyrophosphate]	Oxidative decarboxylation & transketolase reaction.
2) Riboflavin (B ₂)	FDA & FMN	Oxidation & Reduction reaction.
3) Nicotin (B ₃)	NAD ⁺ & NADP ⁺	Oxidation & Reduction reaction.
4) Pantothenic acid (B ₅)	Coenzyme A	Acyl carrier.
5) Pyridoxine (B ₆)	PLP [Pyridoxal phosphate]	Transamination, Deamination Decarboxylation of amino acid.

6) Biotin [B_7]	Biocytin	Carboxylation reaction.
7) Folic acid [B_9]	THF [Tetrahydro folic acid]	Carrier of 1 carbon group.
8) Cyanocobalamin [B_{12}]	Methylcobalamin & deoxyadenosylcobalamin	Transfer of CH_3 group & isomerisation.

Imp Question :- Name coenzymes of Vitamin B Complex with their corresponding vitamins & their role in metabolism with examples

→ answer is above table of examples of coenzymes.

Question :- Name enzyme requiring metal elements as a cofactor.

→ Refer table 6.4 an enzyme topic in Naik book.

* Enzyme Inhibition :-

- Any substance that can diminish the velocity of an enzyme catalyzed Reaction called Inhibitor.

- enzyme inhibitors interfere with catalysis & inhibition of reactions.

- Enzyme inhibitors include :-
drugs, antibiotics, toxin & antimetabolites, many natural product of the enzyme reactions.

Types of enzyme inhibitors.

1) Reversible inhibitors :-

- Bind enzyme through Non-covalent bond.
- activity of enzyme is restored fully.
- Inhibitor physically removed from system.

2) Irreversible inhibitors :-

- enzyme bind through covalent Bond, tightly.
- cannot released from the system.
- modify the enzyme permanently.

Types of Enzyme inhibition.

- 1) Competitive inhibition
- 2) Non competitive inhibition
- 3) Un competitive inhibition
- 4) Suicide inhibition.

1) Competitive inhibition
Competitive inhibitor is usually similar to the substrate.

Non-competitive inhibition.
Inhibitor structurally different from the substrate.

2) Inhibitor bind to the enzyme at active site i.e. Inhibitor & substrate binds at same site

Inhibitor & substrate may bind at different sites. i.e. Inhibitor & substrate binds at different sites.

3) V_{max} remains unchanged increases K_m value.

Induced V_{max} without changing K_m value.

4) Inhibition is reversed by increasing substrate concentration.

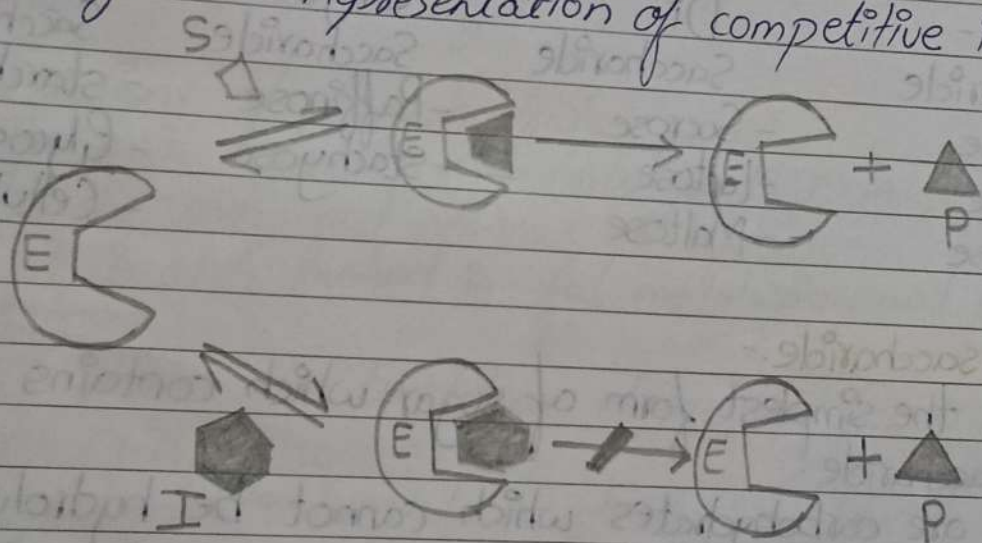
Inhibition cannot overcome by increasing substrate concentration.

5) Competition between substrate & inhibitors because of same binding site.

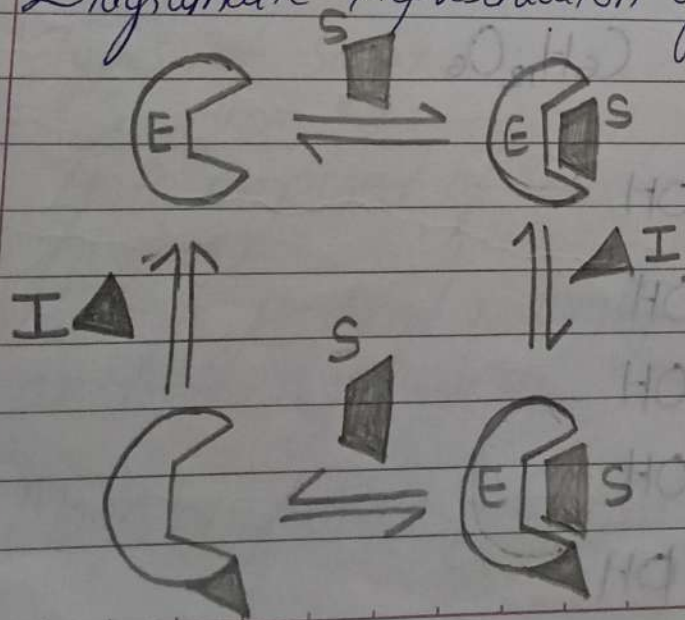
No competition between substrate & inhibitors because different binding sites.

6) eg. Succinate dehydrogenase e.g. ethanol or certain Narcotic drugs of acid cycle & inhibited by malonate which resembles succinate.

• Diagrammatic Representation of competitive inhibition.



• Diagrammatic Representation of Non-competitive inhibition:-



Carbohydrates.

- Carbohydrates are polyhydroxyaldehyde or polyhydroxy ketones of their derivatives.
- Carbohydrates are made up of carbon, hydrogen, oxygen, nitrogen & sulphur.
- Their molecular weight ranges from less than 100-1 million.

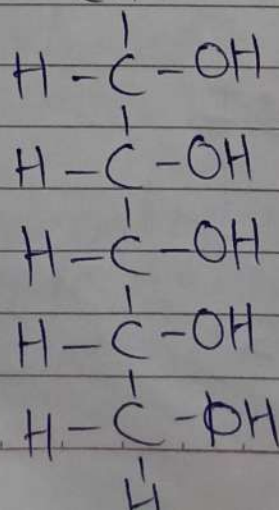
* Classifications of Carbohydrate

Carbohydrate			
Mono-	Di-	Oligo-	Poly-
Saccharide	Saccharide	Saccharide	Saccharide.
- Glucose	- Sucrose	- Raffinose	- starch
- Fructose	- Lactose	- Stachyose	- Glycogen
- Galactose	- Maltose		- Cellulose

Monosaccharide:-

- It is the simplest form of sugar which contains only one saccharide.
- These are carbohydrates which cannot be hydrolysed further into simple form.

Structure of glucose:- $C_6H_{12}O_6$



Disaccharide:-

- i) They are generally made up of 2 sugar units.
- ii) These are carbohydrates which on hydrolysis yield two molecules of same or different monosaccharide.

Oligosaccharide:-

- Carbohydrates formed by condensation of 2 to 9 monomers are called Oligosaccharide.

Polysaccharide:-

- It is the most abundant type of carbohydrate formed by linking simple sugars.

* Functions of Carbohydrate:-

- i) The main function of Carbohydrate is to provide energy to the body and nervous system.
- ii) It is also involved in fat metabolism and prevents ketoses.
- iii) It inhibits the breakdown of proteins for energy as they are primary source.
- iv) Carbohydrate breakdown starch into glucose by the help of amylase.

* Types of Sugar:-

- i) Sucrose
 - It is composed of 2 monosaccharide (glucose and fructose).
 - It is produced naturally in plants and is referred as table sugar and cane sugar.

Molecular formula:- $C_{12}H_{22}O_{11}$

- It is a type of non-reducing sugar.

2) Maltose

- It is composed of 2 glucose units
- wheat, corn, barley contains maltose.
- Maltose has the property to raise blood sugar level.

3) Lactose

- It is formed by secretory cells of mammary glands during lactation.
- It consist of glucose and galactose and they are linked to each other.

4) Glycogen

- It is made up of D-glucose unit it is mostly stored in liver and muscles and is the most important substance in the body.

* Glycosidic bond.

- i) A glycosidic bond is present in disaccharide and polysaccharide.
- ii) This is formed between two adjacent monosaccharides.
- iii) Like peptide bond elimination of water molecules accompanies the formation of glycosidic bond.
- iv) This is dehydration reaction.

* Formation of Glycosidic bond.

It is formed by dehydration or condensation reaction.

Types:-

- i) O-glycosidic bond:- formed between the anomeric

carbon of sugar and a hydroxyl group

- 2) N-glycosidic bond :- A nitrogen atom connects the sugar to another molecule, formed when the hydroxyl group of sugar molecule joins with an amine group
- 3) C-glycosidic bond :- Carbon atom connects to sugar of another molecule.

★ Glycosaminoglycan (GAG)

- 1) Gags are polysaccharides that are present in the body and play a role in cell signalling, from a wound repair, injuries, etc.
- 2) They are found in the extracellular matrix (ECM) surrounding cells.

Examples of glycosaminoglycan :-

- 1) chondroitin sulfate - It is a natural gag found in cartilage, skin, blood vessels, ligaments, tendons, brain, etc.
- 2) Heparin sulfate - A linear anionic polysaccharide that is closely related to clotting or coagulation
- 3) Hyaluronic acid - A non-sulfated gag found in cartilage, tendon & synovial fluid.
- 4) Keratin sulfate - It regulates cellular function

Mucopolysaccharide.

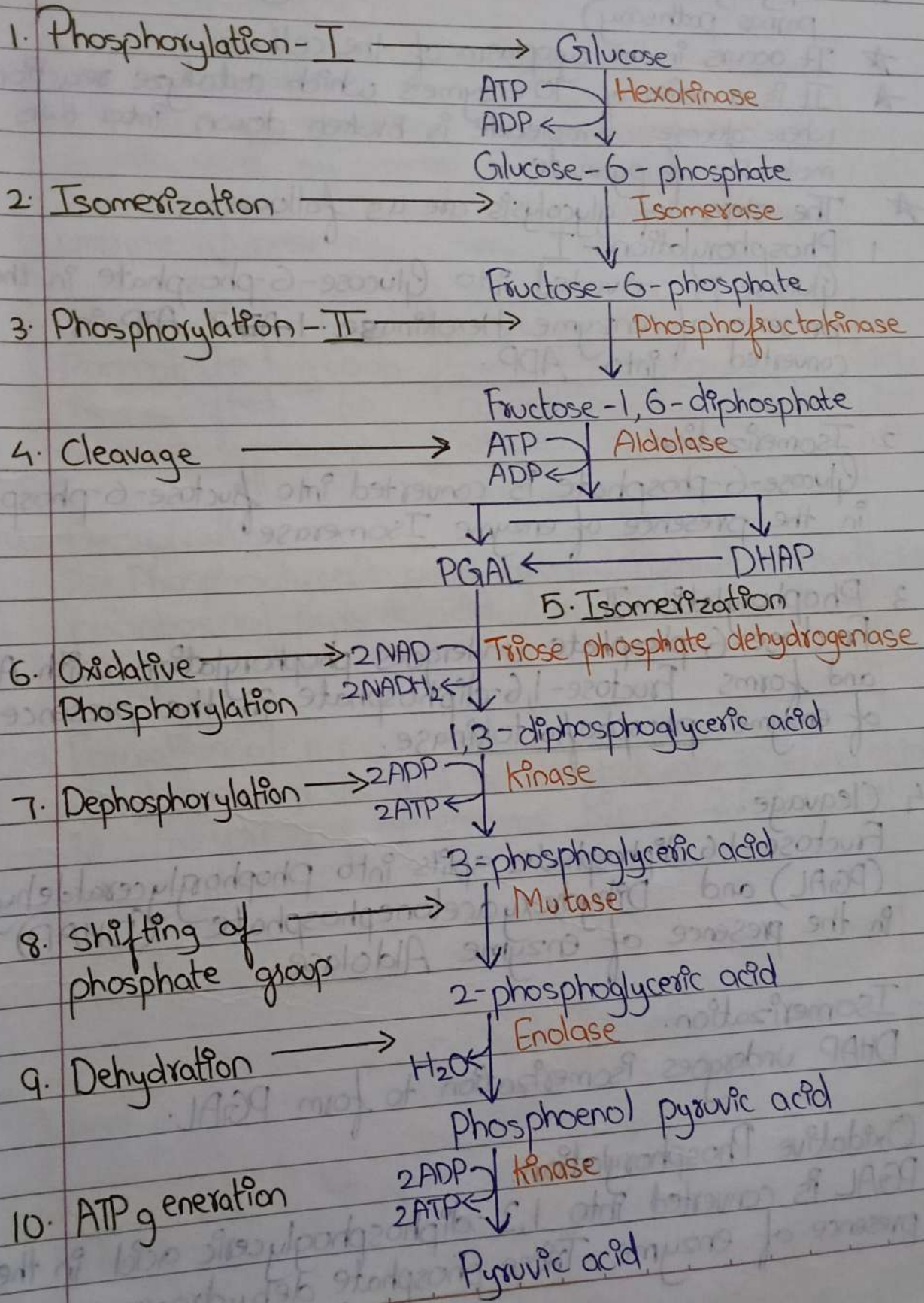
Carbohydrate Metabolism

Def:- Metabolism is a process in which number of biomolecules changes occurs in the body and helps in the exchange of matter and energy between cell and the environment.

Importance of Metabolism.

- 1) Chemical energy is obtained
- 2) Dietary nutrients are used for synthesis of new molecules.
- 3) New molecules unite to form protein, nucleic acid, etc.

Glycolysis



- ★ Glycolysis is known as EMP pathway (Emden mayerhoff parnas pathway)
- ★ It occurs in the cytoplasm of the cell.
- ★ It is a series of 10 enzymes which catalyze reactions where glucose molecule is broken down into two molecules of pyruvate.
- ★ The steps of glycolysis are as follows:-
 1. Phosphorylation-I
Glucose is converted into Glucose-6-phosphate in the presence of enzyme Hexokinase. ~~1-ATP~~ ATP is converted into ADP.
 2. Isomerization.
Glucose-6-phosphate is converted into fructose-6-phosphate in the presence of enzyme Isomerase.
 3. Phosphorylation-II.
Fructose-6-phosphate undergoes phosphorylation with ATP and forms Fructose-1,6-diphosphate in the presence of enzyme phosphofructokinase.
 4. Cleavage.
Fructose-1,6 diphosphate splits into phosphoglyceraldehyde (PGAL) and Dihydroxyacetonephosphate (DHAP) in the presence of enzyme Aldolase.
 5. Isomerization.
DHAP undergoes Isomerization to form PGAL.
 6. Oxidative Phosphorylation.
PGAL is converted into 1,3-diphosphoglyceric acid in the presence of enzyme Triose phosphate dehydrogenase.

2 NAD is converted 2 into 2 NADH₂ (Nicotinamide adenine dinucleotide).

7. Dephosphorylation.

1,3-diphosphoglyceric acid is converted into 3-phosphoglyceric acid by removal of 1 phosphate group. 2 ADP are converted into 2 ATP by the presence of enzyme kinase.

8. Shifting of phosphate group.

Phosphate group from 3-phosphoglyceric acid is shifted to carbon no. 2 in the presence of enzyme Mutase.

9. Dehydration

2-Phosphoglyceric acid after dehydration produces phosphoenol pyruvic acid in the presence of enzyme enolase.

10. Formation of pyruvic acid.

Phosphoenol pyruvic acid is converted into pyruvic acid in the presence of enzyme kinase. 2 ATP molecules are synthesized.

Energetics.

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Reaction	ATP Formed
PGAL to 1,3-Diphosphoglycerate (2NADH_2)	06
1,3-Diphosphoglycerate to 3-phosphoglycerate ($2\text{ADP} \rightarrow 2\text{ATP}$)	02
Pepa to Pyruvate ($2\text{ADP} \rightarrow 2\text{ATP}$)	02
Total	10

Reaction	ATP Formed
Glucose to glucose-6-phosphate ($\text{ATP} \rightarrow \text{ADP}$)	01
Fructose-6-phosphate to Fructose 1,6-Diphosphate ($\text{ATP} \rightarrow \text{ADP}$)	01
Total	02

Net ATP acquired = ATP formed - ATP consumed.

Net ATP acquired = $10 - 2$

Therefore, Net ATP acquired = 8 ATP.

$\therefore$ In the reaction of glycolysis where glucose is converted Pyruvic acid and Net ATP acquired is 8.

Def:- Glycolysis is a pathway of carbohydrate metabolism where glucose is catabolized into pyruvic acid.

- 1) Glycolysis takes place in aerobic condition where the by product is pyruvic acid.
- 2) When glucose is converted into lactic acid, this condition is anaerobic.

Difference between aerobic & anaerobic

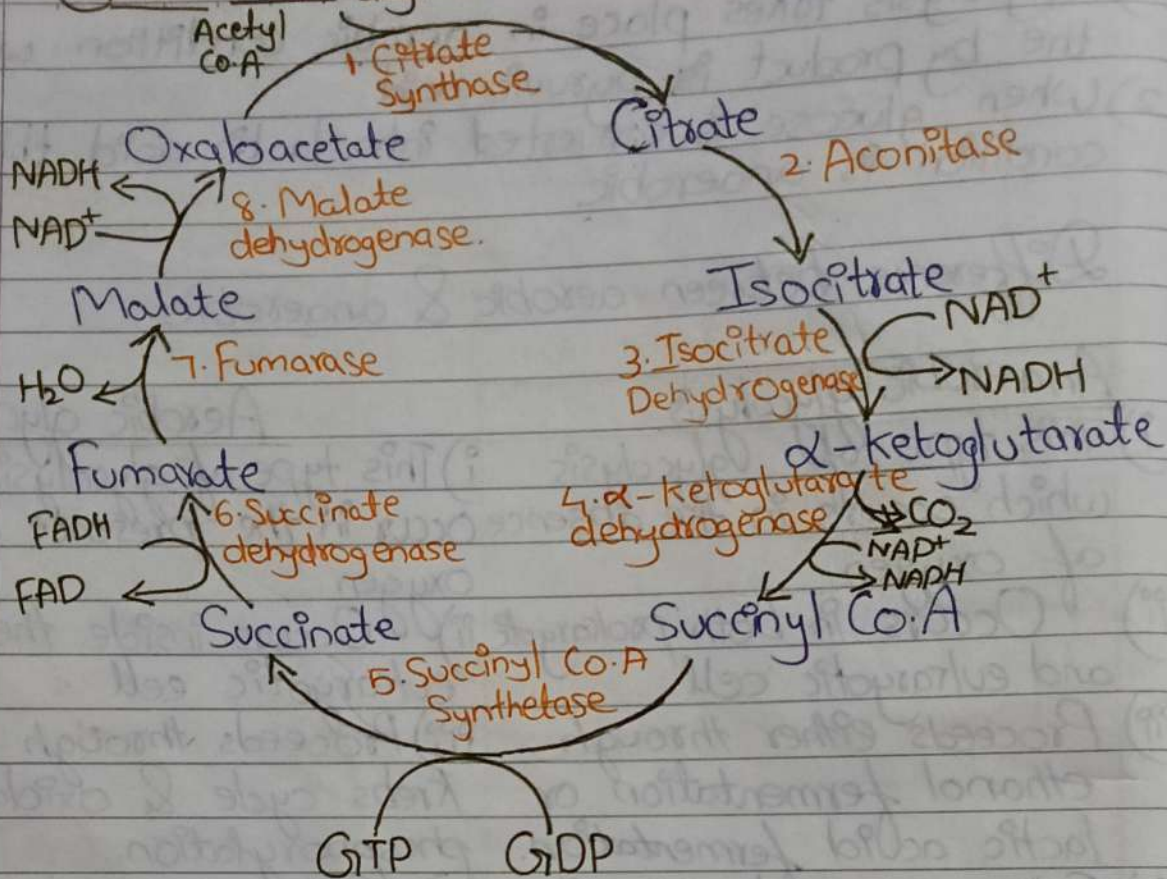
Anaerobic glycolysis

- i) This type of glycolysis which occurs in the absence of oxygen.
- ii) Occurs in both prokaryotic and eukaryotic cell.
- iii) Proceeds either through ethanol fermentation or lactic acid fermentation.
- iv) Further proceeds into the cytosol.
- v) Responsible for reducing pyruvate into lactate in the lactic acid fermentation or acetaldehyde in the ethanol fermentation.
- vi) Lactic acid or ethanol.
- vii) Co-factors do not involve production of ATP.
- viii) Leads to a less efficient ATP production pathway which produces 2 ATPs per glucose molecule.

Aerobic glycolysis

- i) This type of glycolysis which occurs in the presence of oxygen.
- ii) Occurs inside the eukaryotic cell.
- iii) Proceeds through the Krebs cycle & oxidative phosphorylation.
- iv) Further proceeds inside the mitochondria.
- v) Responsible for oxidizing pyruvate into acetyl Co-enzyme A.
- vi) End product carbon dioxide & water.
- vii) Co-factors involve the production of ATP.
- viii) Leads to a significantly efficient ATP production pathway which produces 32 ATPs per glucose molecule.

Citric Acid Cycle.



- 1) The kreb cycle or TCA (Tricarboxylic acid cycle) or Citric acid cycle is a series of enzyme catalyzed reactions occurring into the mitochondrial matrix where acetyl Co. A is further oxidized to form Carbondioxide. & co-en-zymes are reduced. which generate ATP into the electron transport chain (ETC)
- 2) The first product of this cycle ~~that is~~ i.e citric acid contains three carboxyl groups hence it is known as Tricarboxylic acid cycle.
- 3) It is a series of 8 step process where Acetyl group of Acetyl Co. A is oxidized to form 2 molecules of carbondioxide and 1 ATP is produced
- 4) It is an aerobic process which requires oxygen as the final oxidant of the reduced coenzymes.

5) Reduced high energy compounds NADH (Nicotinamide adenine dinucleotide + H⁺) and FADH₂ (Flavin adenine dinucleotide + H⁺) are also produced.

6) STEP 1:-

Condensation of Acetyl Co.A with four carbon compound (Oxaloacetate) to form six carbon citrate Co.A is released. this reaction is catalyzed by Citrate Synthase.

7) STEP 2:-

Citrate is converted into Isocitrate (which is an isomer) this reaction is catalyzed by enzyme Aconitase.

8) STEP 3:-

Isocitrate undergoes dehydrogenation and decarboxylation to form five carbon α -ketoglutarate. A molecular form of carbon dioxide is released. Isocitrate dehydrogenase catalyzes the reaction. NAD⁺ is converted to NADH.

9) STEP 4:-

α -ketoglutarate undergoes oxidative decarboxylation to form Succinyl Co.A, a 4 Carbon compound. This reaction is catalyzed by α -ketoglutarate dehydrogenase enzyme complex. 1 molecule of CO₂ is released and NAD⁺ is converted to NADH.

10) STEP 5:-

Succinyl Co.A forms succinate this enzyme Succinyl Co.A synthetase catalyze the reaction. This is coupled by substrate level phosphorylation of

of GDP (Guanosine diphosphate) to get GTP. GTP transfers its phosphate to ADP forming ATP.

11) STEP 6:-

Succinate is oxidized by the enzyme Succinate dehydrogenase to Fumarate. FAD is converted into FADH.

12) STEP 7:-

Fumarate gets converted to Malate by the addition of 1 H_2O . The enzyme catalyzing this reaction is Fumarase.

13) STEP 8:-

Malate is dehydrogenated to form Oxaloacetate which combines with another molecule of Acetyl Co.A and starts the new cycle. Hydrogens removed gets transferred to NAD^+ forming NADH. This reaction is catalyzed by Malate dehydrogenase.

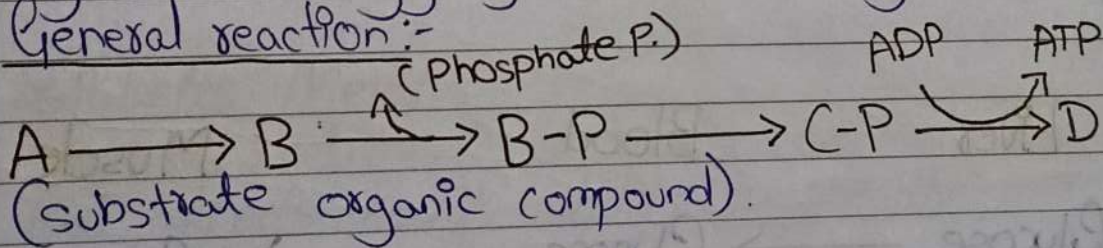
* Substrate level Phosphorylation.

1. Phosphorylation involves the transfer of phosphate group from one compound to other.
2. Substrate level phosphorylation is a direct phosphorylation ADP with a phosphate group by using the energy obtained from a coupled reaction.
3. It occurs in Cytoplasm (Glycolysis) due to aerobic and anaerobic condition, and in mitochondrial matrix (Kreb cycle - anaerobic condition).
4. The Synthesis of ATP by the direct transfer of phosphate group from a substrate which is high energy intermediate to a molecule of ADP.

★ Mechanism of Substrate level Phosphorylation.

1. High energy phosphate group are transferred directly from phosphorylated substrate to ADP.
2. ATP is synthesized via substrate level phosphorylation in glycolysis and kreb cycle.

General reaction:-



* Glycogen Metabolism.

① Glycogenesis.

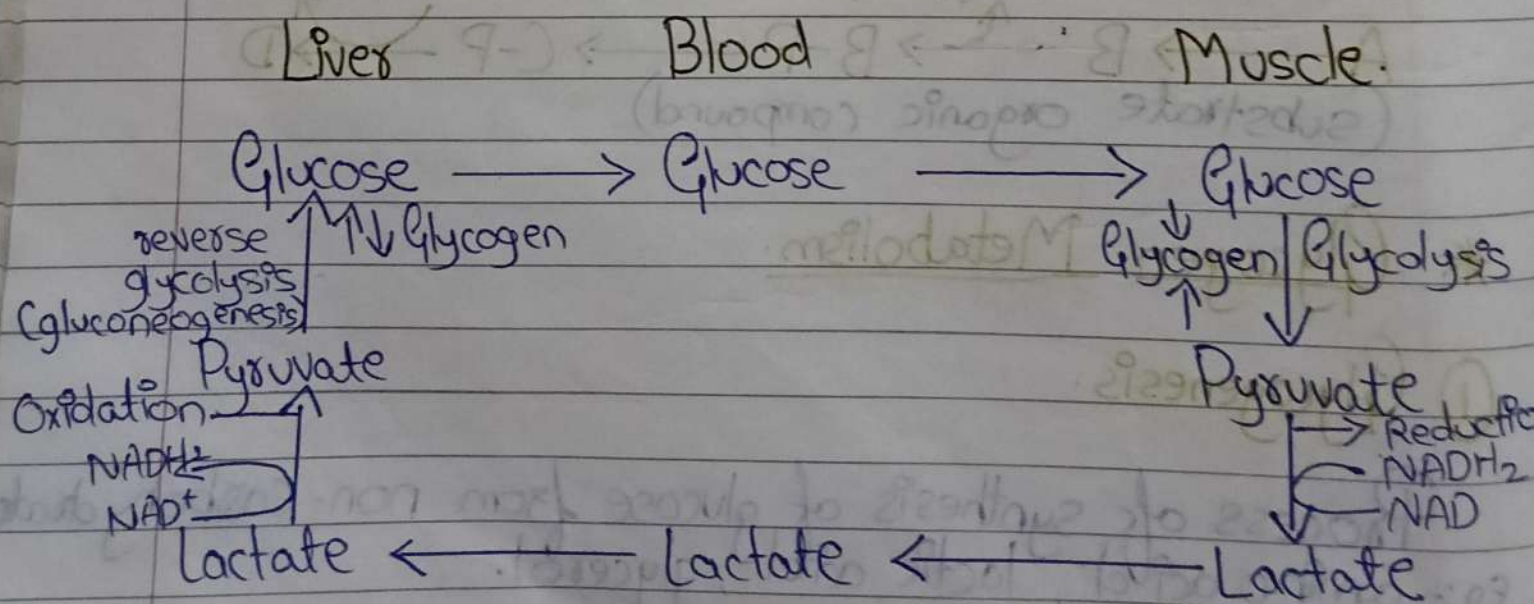
Process of synthesis of glucose from non-carbohydrate
eg:- amino acid, lactic acid, glycerol.

② Glycogenolysis

- i) This process is referred as breakdown of glycogen in the tissues it may be broken down to glucose in liver and kidneys or glucose-6-phosphate in muscles.
- ii) The process gets enhanced by hypoglycemia (when glucose level in the blood is low) or under the influence of certain hyperglycemic hormones.
- iii) Liver glycogen is metabolically more easily available for this pathway as compared to muscle glycogen.

Cori Cycle

* Karl Cori introduced the cycle of conversion of lactate to glucose in the liver which release glucose into the blood and uptake of glucose by peripheral tissue, conversion of glucose to lactate by glycolysis by release of lactate into the blood and uptake of lactate by the liver for conversion again into the glucose.



i) The Pyruvate formed in muscle by degradation of glycogen or glucose is oxidized into CO_2 and H_2O in the muscle itself when oxygen supply is sufficient.

ii) Pyruvate is reduced to lactate which enters the blood and is taken up by liver where it is converted to glucose and glycogen.

iii) This cycle is also known as lactic acid cycle.

iv) Glucose / Glycogen is converted $\longrightarrow$ lactate

Glucose / Glycogen $\longrightarrow$ lactate
Muscle

v) Lactate $\longrightarrow$ Glucose
Liver

vi) Conversion of muscle lactate to glucose in liver and its re-entry into muscle is called Cori cycle.

Glucose $\xrightleftharpoons[\text{Liver}]{\text{muscle}}$ Lactate

Diabetes Mellitus.

i) Diabetes Mellitus is a chronic condition or a disease that affects body's ability to use the energy found in food.

ii) It is a condition where pancreatic glands does not generate enough insulin required by the body to regulate glucose metabolism.

iii) The body breaks down carbohydrate & sugar present in food and then converts them into special sugar called glucose.

iv) This glucose is used to fuel up the body, excess

- glucose present in the blood is converted into glycogen as an energy reserve this process requires insulin
- v) Diabetes mellitus is a condition during which body does not produce sufficient insulin which leads to high blood glucose levels
 - vi) This condition relates to pancreas, an organ which involves production of insulin.
 - vii) It is a disease in which one finds raised blood sugar level (hyperglycemia)

★ Main 3 symptoms of diabetes are:-

- 1) Polydipsia.
- 'Increased thirst' (a person feels thirsty and frequency of water intake increases).
- 2) Polyphagia.
- Condition in which a person feels hungry than the usual one.
- 3) Polyuria.
- Frequency of urination is increased

Types of diabetes:-

- 1) Type I diabetes
'A chronic condition in which the pancreas produces little or no insulin'
- 2) Type II diabetes
'A chronic condition in which the body is capable to process the sugar but with difficulty, it may be treated with life style changes, physical exercise and oral anti-diabetic drugs. Generally it takes

place above 40 years of age.

It is called maturity on set age.

In this type insulin is secreted but not as much as it should.

3) Pre-diabetes.

'A condition in which blood sugar is high but not high enough to be type-II diabetes.'

4) Gestational diabetes.

'A form of high blood sugar affecting pregnant woman only.' Such woman become normal once the delivery has taken place but they are on risk after the age of 40 years or so.

5) Renal diabetes.

In this condition the blood glucose level are normal but sugar appears in urine. This is due to very low kidney threshold for glucose.

Water And Electrolytes.

Insufficient H_2O
in the body

Decreases blood
volume

Increases blood
osmolarity

Decreases blood
pressure

Osmoreceptors in
the hypothalamus

Dry mouth

Increases
angiotensin II

Stimulates thirst
center in the hypothalamus

Increases thirst

H_2O taken in

Decreases blood
osmolarity

Water Balance

Body water Distribution: Water is the largest component of the human body, accounting for approximately 55-70% of total body weight.

- Water is distributed between intracellular fluid (ICF) and (EEF) Extra cellular fluid compartments.

20 10 10

Electrolytes

- The ICF, inside cells, contains about 2/3rds of the body's water.
- The ECF, outside cells; includes plasma and interstitial fluid.
- Total body water (TBW) varies based on factors like age, sex and body composition
eg:- more fat means less water.

Water Turnover:

- Water turnover refers to the continuous exchange of water between the body and the environment.
- External balance involves Input and Output.
- Internal fluxes refer to the movement of water between body compartments.
- Maintaining a balance between water input and output is crucial for proper body function.

Regulation of Water Balance:

- ADH (Antidiuretic hormone) or Vasopressin:
- ADH is produced by the hypothalamus and released by the pituitary gland.
- It acts on the kidneys to increase water reabsorption thus reducing urine volume and concentrating urine.
- ADH release is triggered by increased blood osmolarity or decreased blood volume.
- This mechanism helps conserve water and maintain blood volume and osmolarity.

Thirst Center:

- The thirst center is located in the hypothalamus.
- It senses changes in blood osmolarity and volume.

- when blood osmolarity increases or blood volume decreases, the thirst center stimulates the sensation of thirst, prompting fluid intake.
- Drinking fluids helps restore water balance by replenishing lost water and diluting the blood.

Biochemistry of Regulation:

- **Osmoreceptors:** Specialized cells in the hypothalamus detect changes in blood.
- **Baroreceptors:** These receptors in blood vessels and the kidneys detect changes in blood pressure and volume.
- **Renin-Angiotensin-Aldosterone-System (RAAS):** This system is also involved in regulating blood volume and pressure, and can influence ADH release.
- **ADH receptors:** ADH binds to receptors on kidney cells, triggering the insertion of aquaporins (water channels) into the collecting duct walls, allowing water to be reabsorbed.

Electrolyte Balance

The osmolarity of body fluids and electrolyte distribution are tightly regulated by hormones like:

1. Aldosterone
2. Renin-angiotensin-aldosterone-system

3. Atrial natriuretic factor (ANF).

These systems work to maintain fluid and electrolyte balance, which is crucial for proper cell function and overall homeostasis.

Osmolarity and Electrolyte Distribution:

- **Osmolarity**: refers to the concentration of solute particles (like electrolytes) in a solution. In the body, it's a key factor in determining fluid movement between intracellular and extracellular compartments.
- **Electrolytes**: like sodium, potassium, chloride, calcium and magnesium play vital roles in various bodily functions, including nerve and muscle function, fluid balance and acid-base balance.
- **Sodium**: is the primary electrolyte in the extracellular fluid, while potassium is the major intracellular electrolyte.
- **Water**: moves across cell membranes in response to osmotic gradients created by differences in electrolyte concentrations.
- **Role of Aldosterone**:
 - Aldosterone is a hormone produced by the adrenal glands that plays a crucial role in regulating sodium and potassium balance.

- It acts on the kidneys to increase sodium reabsorption in the distal convoluted tubules and collecting ducts, and also promotes potassium excretion.
- By influencing sodium and potassium levels, aldosterone indirectly regulates water balance, as water tends to follow the movement of sodium.
- Aldosterone is a key player in the RAAS, helping to maintain blood and fluid balance.

Renin-Angiotensin-Aldosterone-System (RAAS):

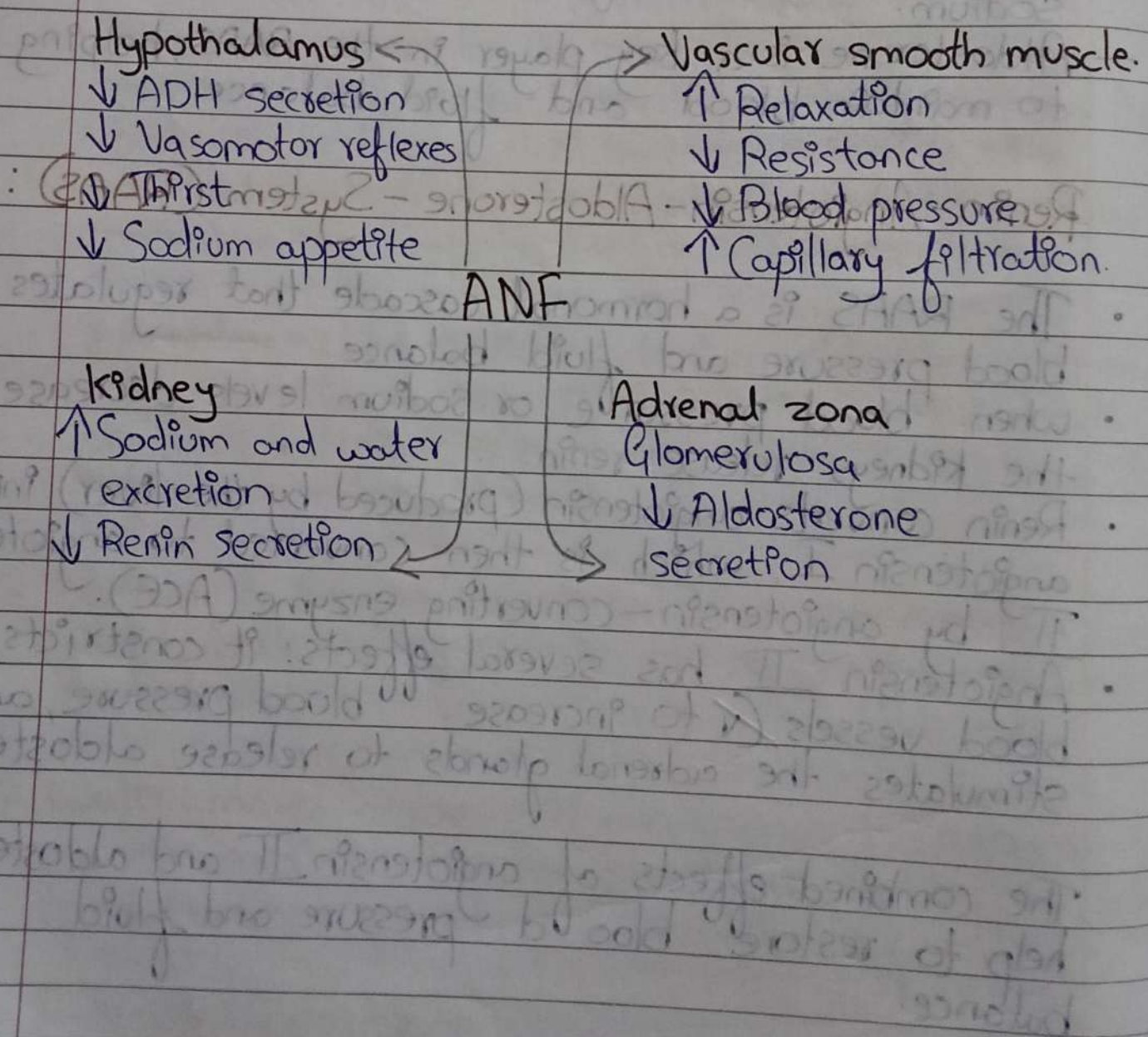
- The RAAS is a hormonal cascade that regulates blood pressure and fluid balance.
- When blood pressure or sodium levels decrease, the kidneys release renin.
- Renin converts angiotensin (produced by the liver) into angiotensin I, which is then converted to angiotensin II by angiotensin-converting enzyme (ACE).
- Angiotensin II has several effects: it constricts blood vessels to increase blood pressure, and stimulates the adrenal glands to release aldosterone.

The combined effects of angiotensin II and aldosterone help to restore blood pressure and fluid balance.

Role of Atrial Natriuretic Factor (ANF):

- ANF is a hormone produced by the heart that counteracts the effects of the RAAS.

- When blood volume or pressure increases, the heart release ANF.
- ANF promotes sodium and water excretion by the kidneys, helping to reduce blood volume and pressure.
- ANF acts a counter-regulatory hormone to the RAAS, promoting fluid and electrolyte balance in the opposite direction.



Role of Atrial Natriuretic Factor (ANF):

Muscle Contraction Mechanism

* Muscle Contraction

- Muscle contraction relies on the interaction of contractile proteins, primarily actin and myosin, within muscle fibers.
- This process involves a series of biochemical reactions that convert chemical energy (from ATP) into mechanical energy, resulting in muscle shortening.
- The steps include nerve stimulation, calcium release, cross-bridge cycling between actin and myosin and ATP hydrolysis.

Contractile Elements:

- **Actin:** A protein that forms the thin filaments in muscle fibers. It has binding sites for myosin.
- **Myosin:** A protein that forms thick filaments. It has a head region that can bind to actin and hydrolyze ATP.
- **Troponin and Tropomyosin:** Regulatory proteins associated with actin filaments. They control the interaction between actin and myosin.
- **Sarcomere:** The basic unit of muscle contraction, composed of repeating units of actin and myosin filaments.

Process of Muscle Contraction:

1. **Nerve stimulation:** An action potential travels down a motor neuron and triggers the release of the neurotransmitter acetylcholine at the neuromuscular

junction.

2. **Calcium Release:** Acetylcholine initiates an action potential in the muscle fiber, causing the sarcoplasmic reticulum (SR) to release calcium ions.

3. **Cross-Bridge formation:** Calcium binds to troponin, causing a conformational change that exposes the myosin-binding sites on actin. Myosin heads then bind to actin, forming cross-bridges.

4. **Sliding of filament:** Myosin heads change shape (power stroke) pulling the actin filaments towards the center of the sarcomere.

5. **ATP binding & detachment:** A new ATP molecule binds to the myosin head, causing it to detach from the actin filament.

6. **Hydrolysis and Recocking:** ATP is hydrolyzed to ADP and inorganic phosphate, which cocks the myosin head back into a high-energy position ready to bind to another actin site.

7. **Cycle Repetition:** Steps 3-6 repeat as long as calcium ions are present and ATP is available, causing the filaments to slide past each other and the muscle to contract.

8. **Muscle Relaxation:** When the nerve signal stops, calcium ions are pumped back into the SR, and the troponin-tropomyosin complex covers the myosin-

binding sites on actin, preventing further cross-bridge formation and the muscle relaxes.

Energy for Muscle Contraction:

- **ATP hydrolysis:** The primary energy source for muscle contraction is ATP. The energy released during ATP hydrolysis powers the conformational changes in myosin and the movement of the filaments.
- **Creatine Phosphate:** A molecule that can quickly donate a phosphate group to ADP to regenerate ATP, providing a rapid source of energy for the initial seconds of muscle contraction.
- **Glycolysis:** The breakdown of glucose to produce ATP, which can occur both with and without oxygen.
- **Aerobic respiration:** The breakdown of glucose and other fuel molecules using oxygen to produce a large amount of ATP, supporting sustained muscle activity.

Digestion and Absorption.

Starch Glycogen

Short branched polysaccharides

limit dextrins

α -Dextrinase

Monosaccharides
glucose

maltose

maltase

2 glucose

Disaccharides.

Sucrose

Sucrase

1 glucose
1 fructose

Lactose

Lactase

1 glucose
1 galactose

Digestion of Carbohydrates

Among carbohydrates, only the monosaccharide forms are absorbed. Hence all carbohydrates must be digested to glucose, galactose and fructose for absorption to proceed.

Enzymes Involved

- α -Amylases (salivary and pancreatic) hydrolyze 1,4-glycosidic bonds in starch, yielding maltose, maltotriose and α -limit dextrins
- Maltase α -dextrinase and sucrase in the intestinal brush border then hydrolyze the oligosaccharides to glucose.
- Lactase, maltase and sucrase degrade their respective

disaccharides lactose, trehalose and sucrose to monosaccharides.

- lactase degrades lactose to glucose and galactose
- maltase degrades maltose to glucose
- Sucrase degrades sucrose to glucose and fructose.

Absorption of Carbohydrates

1. Glucose and Galactose

- They are transported from the intestinal lumen into the cells by a Na^+ -dependent co-transport (SGLT1) in the luminal membrane.
- The sugar is transported "uphill" and Na^+ is transported "downhill".
- They are then transported from cell to blood by facilitated diffusion (GLUT2)
- The Na^+ - K^+ pump in the basolateral membrane keeps the intracellular (Na^+) low, thus maintaining the Na^+ gradient across the luminal membrane.

Proenzymes (zymogens) $\longrightarrow$ Active enzymes.

Pepsinogen $\xrightarrow{\text{H}^+}$ Pepsin

Trypsinogen $\xrightarrow{\text{Enteropeptidase}}$ Trypsin

Chymotrypsinogen $\xrightarrow{\text{Trypsin}}$ Chymotrypsin

Proelastase $\xrightarrow{\text{Trypsin}}$ Elastase

Procarboxypeptidases $\xrightarrow{\text{Trypsin}}$ Carboxypeptidase

* Digestion of proteins

- Dietary proteins are a source of amino acids that are utilized for the formation of various cellular substances.
- Mostly proteins must be broken down into amino acids for absorption.
- Digestive products of protein can be absorbed as amino acids, dipeptides & tripeptides.
- Both endopeptidases enzymes which degrade proteins by hydrolyzing interior peptide bonds.
- Exopeptidases enzymes that hydrolyzes one amino acid at a time of proteins and peptides are involved in the digestion of proteins.
- Digestion takes place in the stomach and the small intestine.

In the stomach:

parietal cells secrete - HCl

chief cells secrete - pepsinogen

pepsinogen $\xrightarrow{HCl}$ pepsin (general protease with preference for acidic & aromatic amino acids)

endopeptidases

Exocrine Pancreas secretion into the small intestine:

- trypsinogen $\xrightarrow{\text{enteropeptidase}}$ trypsin
 - chymotrypsinogen $\xrightarrow{\text{trypsin}}$ chymotrypsin
 - proelastase $\xrightarrow{\text{trypsin}}$ elastase
- a. (arginine, lysine) b. (tryptophan, phenylalanine, tyrosine)

leucine). c. (alanine, glycine, serine)

exopeptidases

procarboxypeptidases $\xrightarrow{\text{trypsin}}$ Carboxypeptidases
 B. (basic amino acids: arginine, lysine).
 A. (hydrophobic amino acids)

Secretion by the brush border of the small intestine:

aminopeptidases (many)

Enzymes involved

- Pepsin
 - Pepsin is secreted in its zymogen form as pepsinogen by the chief cells of the stomach.
 - Pepsinogen is activated to pepsin by gastric H^+ . The optimum pH for pepsin is between 1 and 3.
 - Pepsin hydrolyzes proteins into peptides and proteoses.
 - When the pH is >5 , pepsin is denatured. Thus in the intestine, as HCO_3^- is secreted in pancreatic fluids, duodenal pH increases, and pepsin is inactivated.
- Pancreatic proteases
 - The digestion is completed in the small intestine by the action of pancreatic and intestinal juice.
 - The proteases include trypsin, chymotrypsin, elastase, carboxypeptidase A and B.

- They are secreted in inactive forms that are activated in the small intestine as follows:
 - Trypsinogen is activated to trypsin by a brush border enzyme, enterokinase.
 - Trypsin then converts chymotrypsinogen, pro-elastase and procarboxypeptidase A and B into their active forms.

* Absorption of proteins

1. Free amino acids.

- Na^+ -dependent amino acid cotransport occurs in the luminal membrane. It is analogous to the cotransporter for glucose & galactose.
- The amino acids are then transported from cell to ~~body~~ blood by facilitated diffusion.
- There are 4 separate carriers for neutral, acidic, basic and imino amino acids respectively.

2. Dipeptides & tripeptides.

- They are absorbed faster than free amino acids.
- H^+ -dependent cotransport of dipeptides and tripeptides also occurs in the luminal membrane.
- After the dipeptides & tripeptides are transported into the intestinal cells, cytoplasmic peptidases hydrolyze them to amino acids.
- The amino acids are then transported from cell to blood by facilitated diffusion.

* Digestion of fats.

- Fats not being soluble in water by their nature are both difficult to digest and absorb. They do not mix with the stomach or intestinal contents.
- Lipids include triglycerides, phospholipids, cholesterol, sterols and fat-soluble vitamins.
- The first step in lipid digestion is emulsification, which is the transformation of large lipid droplets into much smaller droplets.
- The emulsification process increases the surface area of the lipid-exposed to the digestive enzymes by decreasing the droplet size.

Enzymes Involved.

1. In the mouth.

- Lingual lipases digest some of the ingested triglycerides to monoglycerides and fatty acids.
- However, most of the ingested lipids are digested in the intestine by pancreatic lipases.

2. Stomach

- In the stomach, mixing breaks lipids into droplets to increase the surface area for digestion by pancreatic enzymes.

3. Small intestine

- Bile acids emulsify lipids in the small intestine increasing the surface area for digestion. The hydrophobic products of lipid digestion are solubilized in micelles by bile acids.
- Pancreatic lipases hydrolyze lipids to fatty acids, monoglycerides, cholesterol. The enzymes are pancreatic lipase, cholesterol ester hydrolase, and

phospholipase A2.

* Absorption of fats.

- Micelles bring the products of lipid digestion into contact with the absorptive surface of the intestinal cells.
- Then, fatty acids, monoglycerides and cholesterol diffuse across the luminal membrane into the cells. Glycerol is hydrophilic and is not contained in the micelles.
- In the intestinal cells, the products of lipid digestion are re-esterified to triglycerides, cholesterol ester and phospholipids, and with apoproteins, form chylomicrons.
- Chylomicrons are transported out of the intestinal cells by exocytosis.
- Because chylomicrons are too large to enter the capillaries, they are transferred to lymph vessels and are added to the bloodstream via the thoracic duct.

Digestive enzymes in the small intestine & pancreas

Pancreas.

- Maltase - digests maltose
- Exopeptidase - removes amino acids until dipeptides are left
- Nuclease - digests RNA & DNA
- Dipeptidase - digests peptides into single amino acids.
- Lactase - digests lactose
- Sucrase - digests sucrose

Small Intestine.

- Lipase - digests triglycerides in fatty acid + glycerol
- Amylase - digests starch into maltose.
- phospholipase - digests phospholipids into fatty acids + glycerol & phosphate grp.
- Insulin - regulates glucose in blood.
- Endopeptidase - digests proteins into short polypeptides.

Clinical Biochemistry

- Clinical biochemistry explores the levels of various substances in blood and urine to assess organ function, particularly the liver and kidneys.
- Normal levels for blood glucose, urea, uric acid, creatinine, calcium, phosphates, pH and bicarbonate are crucial indicators of health and deviations can signal potential issues.

Liver and renal function tests further evaluate the health and efficiency of these vital organs.

* Normal Blood and Urine Constituents:

1. Glucose: Blood glucose is maintained with a narrow range, and its levels are tightly regulated by insulin and glucagon. Abnormal levels can indicate diabetes or other metabolic disorders.
2. Urea: Urea is a waste product of protein breakdown and its levels in blood and urine reflect kidney function. Elevated urea levels can indicate impaired kidney function.
3. Uric Acid: Uric acid is a byproduct of purine metabolism, and its levels can be affected by diet and kidney function. Elevated uric acid levels can lead to gout or kidney stones.
4. Creatinine: Creatinine is a waste product of muscle metabolism, and its levels in blood and urine are

used to assess kidney function. Elevated creatinine levels can indicate kidney damage.

5. Calcium and Phosphates: These minerals are essential for bone health, nerve and muscle function and blood clotting. Their levels in blood and urine are closely regulated and can be affected by kidney and parathyroid function.

6. pH and Bicarbonate: Blood and urine pH reflect the body's acid-base balance. Bicarbonate is a key buffer that helps maintain pH homeostasis. Abnormal levels can indicate respiratory or metabolic disturbances.

* Importance of Blood & levels:

- Glucose: Blood glucose levels are crucial for diagnosing diabetes and monitoring its management. Urine glucose can also be a sign of uncontrolled diabetes.
- Urea, creatinine and Uric Acid: Elevated levels can indicate kidney dysfunction, while low levels may be associated with muscle wasting or liver diseases.
- Calcium & Phosphate: Abnormal levels can be indicative of bone disorders, kidney disease or parathyroid gland dysfunction.

- **pH & Bicarbonate:** These levels help assess the body's acid-base balance, which can be affected by respiratory or metabolic conditions.

- * **Liver Function Tests:** Liver function tests assess the health and functionality of the liver. These tests measure the levels of various enzymes, proteins and other substances in the blood that are produced or affected by the liver.

Common liver function tests include:

- **Aspartate aminotransferase (AST):** Elevated levels can indicate liver damage or disease.
- **Alanine aminotransferase (ALT):** Elevated levels can indicate liver damage or disease.
- **Alkaline phosphatase (ALP):** Elevated levels can indicate liver or biliary tract obstruction.
- **Bilirubin:** Elevated levels can indicate liver disease or impaired bile flow.
- **Albumin:** Decreased levels can indicate liver dysfunction or malnutrition.

- * **Renal Function Tests:** Renal function tests assess the health and functionality of the kidneys. These tests measure the levels of various substances in the blood and urine that are filtered to excreted by the kidneys.

Common renal function tests include:

- **Blood Urea Nitrogen (BUN):** Elevated BUN levels can indicate kidney dysfunction.
- **Creatinine:** Elevated creatinine levels can indicate kidney dysfunction.
- **Glomerular Filtration Rate (GFR):** GFR measures how well the kidneys are filtering waste from the blood. A decreased GFR can indicate kidney disease.
- **Electrolytes:** Electrolyte levels (sodium, potassium, chloride, etc.) in blood and urine are closely monitored to assess kidney function and fluid balance.
- **Urine Analysis:** Urine Analysis, including specific gravity, protein and microscopic examination, provides valuable information about kidney health.

Albunin

Renal function tests

Nutrition.

- Nutrition plays a vital role in maintaining overall health and well-being.
- Adequate nutrition provides the body with the necessary energy, macronutrients and micronutrients to function optimally.
- A well-balanced diet supports growth, development and maintenance of bodily functions, while also reducing the risk of chronic diseases.

* Importance of Nutrition

- Good nutrition is essential for:

1. Energy production
2. Growth & development
3. Maintenance of bodily functions
4. Prevention of chronic diseases (e.g. heart disease, diabetes, certain cancers)
5. Supporting immune function.

- Calorific values: Calorific values represent the energy content of foods. The three main macronutrients and their approximate calorific values are:

1. Carbohydrates: 4 kcal/g
2. Protein: 4 kcal/g
3. Fat: 9 kcal/g

- Respiratory Quotient (RQ): RQ is the ratio of carbon dioxide produced to oxygen consumed during cellular respiration. It indicates the type of fuel

being metabolized.

1. RQ values : i) Carbohydrates : 1.0
 ii) Fat : 0.7
 iii) Protein : 0.8

2. Significance: RQ helps determine the body's energy substrate utilization and can be used to estimate energy expenditure.

- Energy requirement of a person: The energy requirement varies depending on factors like age, sex, weight, height and activity level. A person's energy needs can be estimated using

1. Basal Metabolic Rate (BMR)

2. Activity Multipliers (eg. sedentary, lightly active, moderately active)

- BMR: It is the energy expended at rest in a fasting state and in a thermoneutral environment. It represents the minimum energy required to maintain basic bodily functions.

1. Normal values:

- Adult men: approximately 1,600-2,000 kcal/day.
- Adult women: approximately 1,400-1,800 kcal/day.

2. Factors affecting BMR:

- Age: During periods of rapid growth, the BMR increases by 15-20 per cent because the growth hormone stimulates cell metabolism and new

cells are formed. It is high during infancy, pre-school years and puberty. During pregnancy and lactation it rapidly increases once again. The BMR gradually declines with age at the rate of 2 per cent for each decade after the age of 21 years.

- Sex: The BMR is 10 per cent higher in men as compared to women. The difference in BMR is attributed to a higher proportion of adipose tissue in females and hormonal variations between the sexes.

- Body composition: The amount of muscle tissue and adipose or fatty tissue in the body affects the BMR. An athlete with well-built and little body fat has a higher BMR than a non-athlete with more body fat of the same weight. The metabolic activity in muscle tissue or lean tissue is much more as compared to adipose or fatty tissue.

- Hormones: Disorders of the thyroid gland markedly influence the BMR. Hyperthyroidism, a condition of excessive production of thyroid hormone, increases BMR and hypothyroidism or decreased production of thyroid hormone decreases BMR.

- State of health: The BMR is low during starvation and malnutrition because of reduction in muscle tissue. In diseases and conditions where catabolic processes are high, such as cancers, tuberculosis and burns, BMR is high.

* Recommended Dietary allowance

- The recommended dietary allowance (RDA) is the guideline stating the amount of nutrients to be actually consumed in order to meet the requirements of the body.
- The RDA is based on requirements.
- The requirement for a particular nutrient is the minimum level that needs to be consumed to perform specific functions in the body and to prevent deficiency symptoms.
- It should also maintain satisfactory stores of the nutrients in the body.
- Recommended dietary allowances are based on a person's requirements for different nutrients. In other words,
- Recommended dietary allowances = Requirement + Margin of safety.

* The margin of safety is added to take care of factors such as:

1. Losses during cooking and processing
2. Short periods of deficient intake.
3. Nature of the diet.
4. Individual variations in requirements.

* Balanced diet

- i) A balanced diet is one which includes a variety of foods in adequate amounts and correct proportions to meet the day's requirements of all essential nutrients such as proteins, carbohydrates, fats, vitamins, minerals, water & fiber.

ii) Such a diet helps to promote and preserve good health and also provides a safety margin or reserve of nutrients to withstand short durations of energy.

- A balanced diet takes care of the following aspects.
- 1. It includes a variety of food items.
- 2. It meets the RDA for all nutrients.
- 3. Nutrients are included in correct proportions.
- 4. Provides a safety margin for nutrients.
- 5. It promotes and preserves good health.
- 6. It helps maintain acceptable body weight for height.

* Balanced diet

i) A balanced diet is one which includes a variety of foods in adequate amounts and correct proportions to meet the body's requirements of all essential nutrients such as proteins, carbohydrates, fats, vitamins, minerals, water & fibre.

- 1. Losses during cooking and processing
- 2. Short periods of deficient intake
- 3. Nature of the diet
- 4. Individual variations in requirements

* The margin of safety is added to take care of factors such as

Recommended dietary allowances = Requirement + Margin of safety

Recommended dietary allowances are based on a person's requirements for different nutrients in other words

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Carbohydrates.

Definition: Carbohydrates are a major source of energy for humans providing approximately 45% to 80% of the total caloric intake in different income groups.

* Classification of Carbohydrates

- Carbohydrates, which are of importance in the diet, are classified on the basis of number of sugar units present in them. They may also be classified as:
- Available carbohydrates: Carbohydrates which can be digested in the human body and yield energy when they are oxidized in the body.
- Unavailable carbohydrates: Carbohydrates which cannot be digested because the human body does not contain the enzymes necessary for their breakdown. Unavailable carbohydrates do not provide any energy to the body but are necessary as they perform some important functions in the body such as regular elimination of faecal waste.

* Functions

- They have many functions in the human body.
- i) The chief function of the carbohydrate is to provide energy to the body, to carry basic functions and maintain body temperature.

- ii) **Energy:** All carbohydrates except fibre provide 4 kcal/g of energy and it is the cheapest source of energy available.
- iii) Glucose is the only form of energy used by the CNS when blood glucose level falls, the brain does not receive energy and convulsions may occur.
- iv) Carbohydrates spare proteins from being broken down for energy and are used for body building and repair. In carbohydrate deficient diets, proteins meant for body building and repair are oxidized to meet the most important and primary need of the body, i.e. energy.
- v) They are required for complete oxidation of fat. In a deficiency, fats are broken down rapidly for energy and intermediate products such as ketones are formed in large amounts resulting in a condition called ketosis.
- vi) Carbohydrates can be converted into non-essential amino acids, provided a source of nitrogen is available.

Dietary Carbohydrates

